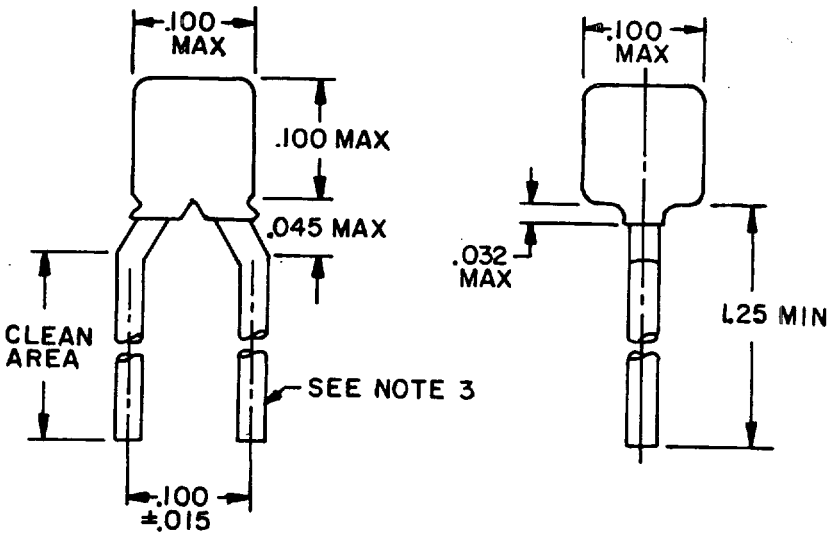
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, CERAMIC DIELECTRIC (TEMPERATURE COMPENSATING), ESTABLISHED AND NON-ESTABLISHED RELIABILITY, STYLES CCR13 AND CC13

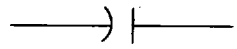
This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the capacitors described herein shall consist of this document and the latest issue of MIL-C-20.

CC13 inactive for new design after 6 June 1984. Use CCR13.



INCHES	MM
.002	.05
.015	.38
.016	.41
.032	.81
.045	1.14
.100	2.54
1.25	31.8



CIRCUIT DIAGRAM

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Lead diameter shall be .016±.002.

FIGURE 1. Styles CCR13 and CC13 capacitors.

Ⓙ denotes changes.

FSC 5910

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Lead type: Radial

Case type: Dipped

DC rated voltage: See table I.

Operating temperature range: -55°C to +125°C.

Characteristics: See table I.

Failure rate level (CCR13 only): M 1.0%), P (0.1%), R (0.01%), or S (0.001%).

Thermal shock and voltage conditioning (CCR13 only): In accordance with MIL-C-20.

Capacitance: Within tolerance specified (see table I).

Dissipation factor: In accordance with MIL-C-20.

Dielectric withstanding voltage: In accordance with MIL-C-20.

Body insulation: Test II.

Temperature coefficient and capacitance drift: In accordance with MIL-C-20. For capacitors with values of 18 pF or less, the temperature-coefficient tolerance shall be determined in accordance with the curve shown on figure 2. For points below the curve, characteristics are not considered practicably measurable.

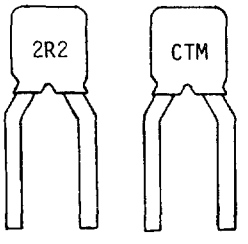
Solderability: In accordance with MIL-C-20.

Resistance to soldering heat: In accordance with MIL-C-20.

Life: In accordance with MIL-C-20, operating condition 2.

Marking: In accordance with MIL-C-20 and method I of MIL-STD-1285 except that minimum marking as shown in the following example will be permitted. Full marking shall be included on the package.

2R2 - Capacitance value.
CMT - Capacitance tolerance (C); FR level, CCR13 only (M) or "-" on CC13; if space permits, manufacturer's trademark, (T).



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Ⓙ TABLE I. Capacitor characteristics.

Type designation 1/ 2/	Rated voltage	Nominal cap.	Cap. tol.	Type designation 1/ 2/	Rated voltage	Nominal cap.	Cap. tol.
	Volts, dc	pF			Volts,dc	pF	
CC-13CHR10B-	100	.10	B	CC-13CH4R7--	100	4.7	BCD
CC-13CHR11B-	"	.11	"	CC-13CH5R1--	"	5.1	"
CC-13CHR12B-	"	.12	"	CC-13CH5R6--	"	5.6	"
CC-13CHR13B-	"	.13	"	CC-13CH6R2--	"	6.2	"
CC-13CHR15B-	"	.15	"	CC-13CH6R8--	"	6.8	"
CC-13CHR16B-	"	.16	"	CC-13CH7R5--	"	7.5	"
CC-13CHR18B-	"	.18	"	CC-13CH8R2--	"	8.2	"
CC-13CHR20B-	"	.20	"	CC-13CH9R1--	"	9.1	"
CC-13CHR22B-	"	.22	"	CC-13CH100--3/	"	10	FGJ
CC-13CHR24B-	"	.24	"	CC-13CG110--	"	11	"
CC-13CHR27B-	"	.27	"	CC-13CG120--	"	12	"
CC-13CHR30B-	"	.30	"	CC-13CG130--	"	13	"
CC-13CHR33B-	"	.33	"	CC-13CG150--	"	15	"
CC-13CHR36B-	"	.36	"	CC-13CG160--	"	16	"
CC-13CHR39B-	"	.39	"	CC-13CG180--	"	18	"
CC-13CHR43B-	"	.43	"	CC-13CG200--	"	20	"
CC-13CHR47B-	"	.47	"	CC-13CG220--	"	22	"
CC-13CHR51B-	"	.51	"	CC-13CG240--	"	24	"
CC-13CHR56B-	"	.56	"	CC-13CG270--	"	27	"
CC-13CHR62B-	"	.62	"	CC-13CG300--	"	30	"
CC-13CHR68B-	"	.68	"	CC-13CG330--	"	33	"
CC-13CHR75B-	"	.75	"	CC-13CG360--	"	36	"
CC-13CHR82B-	"	.82	"	CC-13CG390--	"	39	"
CC-13CHR91B-	"	.91	"	CC-13CG430--	"	43	"
CC-13CH1R0--	"	1.0	BC	CC-13CG470--	"	47	"
CC-13CH1R1--	"	1.1	"	CC-13CG510--	"	51	"
CC-13CH1R2--	"	1.2	"	CC-13CG560--	"	56	"
CC-13CH1R3--	"	1.3	"	CC-13CG620--	"	62	"
CC-13CH1R5--	"	1.5	"	CC-13CG680--	"	68	"
CC-13CH1R6--	"	1.6	"	CC-13CG750--	"	75	"
CC-13CH1R8--	"	1.8	BCD	CC-13CG820--	50	82	"
CC-13CH2R0--	"	2.0	"	CC-13CG910--	"	91	"
CC-13CH2R2--	"	2.2	"	CC-13CG101--	"	100	"
CC-13CH2R4--	"	2.4	"	CC-13CG111--	"	110	"
CC-13CH2R7--	"	2.7	"	CC-13CG121--	"	120	"
CC-13CH3R0--	"	3.0	"	CC-13CG131--	"	130	"
CC-13CH3R3--	"	3.3	"	CC-13CG151--	"	150	"
CC-13CH3R6--	"	3.6	"	CC-13CG161--	"	160	"
CC-13CH3R9--	"	3.9	"	CC-13CG181--	"	180	"
CC-13CH4R3--	"	4.3	"	CC-13CG201--	"	200	"
				CC-13CG221--	"	220	"

1/ Complete type designation will include the following:

1st dash - Symbol "R" (for style CCR13) or dash will be deleted (for style CC13).
2nd dash - Applicable capacitance tolerance symbol.
3rd dash - Applicable failure rate level symbol (CCR13 only) or dash will be deleted (for style CC13).

2/ Capacitors deleted since the previous issue shall be replaced by selecting a capacitor with the nearest parameter.

3/ Capacitance tolerance - ±1% (F) and ±2% (G).

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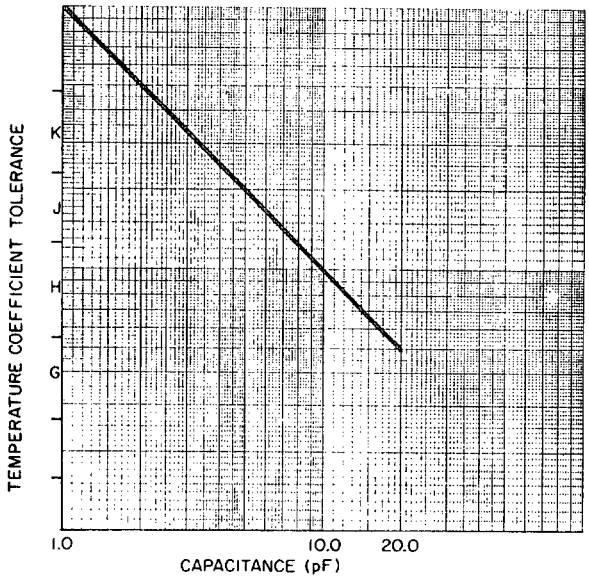


FIGURE 2. Limits for characteristic measurement.

Custodians:
Army - ER
Navy - EC
Air Force - 11

Review activities:
Navy - OS
Air Force - 85, 99
DLA - ES

User activities:
Navy - AS, SH, MC

Agent:
DLA - ES

Preparing activity:
Air Force - 11

(Project 5910-1499)

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